

Amazon Product Analysis

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Background

Amazon is a multinational e-commerce company and has experienced a 41% decline in product purchases during its Prime Day event in 2025 compared to the previous year. As a result, both the *Marketing* and *Sales* departments at Amazon are looking to improve their customer experience. They seek the help of the Lead Data Analyst to provide them with a comprehensive product report which will examine the products carried by Amazon. The *Marketing* department will use this information to deliver product-specific digital promotion while the *Sales* department will use this information to suggest a range of products most suitable for every customer.

Research Questions

A product analysis will be performed for the marketing and sales departments with a focus on answering the following research questions.

Question 1

How does the price, rating and discount compare for products belonging to different categories?

Purpose

Obtain an overview of the products sold by Amazon to make it easier to implement guided marketing and sales strategies.

Impact Hypothesis

When a sales employee is making efforts to sell a product to a future customer, the employee will have sufficient knowledge to answer any questions about the product, including questions relating to pricing and potential discounts.

Question 2

How can the product characteristics be used to predict the product category?

Purpose

Classify new products to ensure appropriate marketing and sales strategies are applied for market success.

Impact Hypothesis

A member of the *Marketing* team is asked to curate digital content to promote a new product. In this case product category classifier models can be used to identify the already existing digital materials prepared for products belonging to the same category.

Question 3

How can product descriptions be used to identify similar products?

Purpose

Match new products to existing products for segmentation and recommendation strategies.

Impact Hypothesis

A sales employee needs to find products similar to the one described by a customer using the product information retrieval models.

Data

The *Amazon Sales Dataset* will be the primary source of data for this project. The dataset contains over 1000 authentic Amazon product records scraped using Python. The dataset contains 16 features, and *Figure 1* provides an overview of the relevant characteristics of the dataset.

Figure 1: Overview of Amazon Sales Dataset

Feature	Description	Type
product_name	String containing product name	Text
category	Pipe separated string containing product categories	Text
discounted_price	Discounted price of product in Indian Rupees (INR)	Numerical with units
actual_price	Price of product in INR	Numerical with units
discount_percentage	Percentage discount on product	Numerical with units
rating	Rating of product from 1 to 5	Ordinal, categorical
rating_count	Number of individuals who voted for a product rating	Numerical
about_product	Description of product	Text

Limitations

- Prices are in Indian Rupees which reduce the interpretability for the North American *Analytics* team.

- No information is provided regarding the purchase history for each product, making it difficult to identify which products require more attention from the marketing and sales departments.

Privacy & Ethics

- The dataset has a Creative Commons (CC) license which provides the necessary permissions to freely use the data for non-commercial purposes.
- The dataset includes the names and user IDs of individuals who contributed to product reviews. Sensitive information such as this does not concern the scope of this project and therefore efforts will be made to ensure this data does not get shared, modified or reproduced in any way.

Analytics Pipeline

Stage 1: Problem Definition

Work with stakeholders from the Marketing and Sales departments to transform their business needs into concise and actionable research questions surrounding Amazon products.

Technologies

Google Docs

Stage 2: Data Preparation and Exploration

Access the data pertaining to the problem context from *Stage 1*, replace any missing values and apply data preprocessing methods to prepare the data for analysis.

Technologies

Python (NumPy, Pandas, SQLite3), Git/GitHub, Google Docs, Google Sheets

Stage 3: Preliminary Analysis

Using the cleaned dataset from *Stage 2*, extract descriptive statistics such as means and frequencies. Compute correlations to observe potential relationships between variables. Visualize variable distributions to observe general trends and outliers using boxplots, histograms, etc.

Technologies

Python (NumPy, Pandas, Seaborn, Matplotlib), Tableau, Git/GitHub, Google Docs

Stage 4: Classification Models

Use the knowledge obtained from *Stage 3* to construct three different product category classifiers: a linear multinomial logistic regression model, and two non-linear classifiers: a

k-nearest neighbours model and a decision tree model. Compare the three models using evaluation metrics such as accuracy, precision and recall.

Technologies

Python (NumPy, Pandas, Sklearn), Git/GitHub, Google Docs

Stage 5: Information Retrieval Models

Use the knowledge obtained from *Stage 3* to construct multiple product information retrieval models using different text preprocessing techniques (stopword removal, lemmatization, etc.). Use TF and TD-IDF statistical weight measures to evaluate importance of words in both unigram and bigram models. Calculate accuracy, precision and recall to evaluate search relevance.

Technologies

Python (NumPy, Pandas, NLTK, Sklearn), Git/GitHub, Google Docs

Stage 6: Interpretations and Recommendations

Summarize the analysis work from the previous stages and produce actionable recommendations to aid decision making processes for stakeholders. The Marketing and Sales departments should be able to utilize the machine learning models and analytical findings to drive product focused business growth. Include comments on code reproducibility and maintenance.

Technologies

Google Docs

GenAI Disclosure

The entirety of this project will be completed without the use of any generative artificial intelligence. Development history of the project will be tracked using GitHub and Google Suite. Justification of all work can be provided upon request.

References

Folk, Z. (2025, July 9). *Amazon's early Prime Day sales dropped 41%, report finds-but Amazon calls that "highly inaccurate."* Forbes.

<https://www.forbes.com/sites/zacharyfolk/2025/07/09/amazons-day-1-prime-day-sales-dropped-41-report-finds/>

J, K. (2023, January 17). *Amazon Sales Dataset*. Kaggle.

<https://www.kaggle.com/datasets/karkavelrajaj/amazon-sales-dataset/data>